

Robust Agency via Constrained Reachability

S. Poole

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Abstract

$$a_t = \nabla \log (\mathbb{E}[R \cdot \Pi] - \lambda \cdot \text{Var}(R \cdot \Pi)) \cdot \Theta$$

We present a constrained reachability control law for agents operating under irreversibility, uncertainty, and adversarial information. Here, R denotes future controllability, Π weights recoverability of future states, λ penalizes outcome variance, and Θ modulates epistemic consistency; external signals are incorporated via adaptive trust weighting.

We derive this expression constructively by identifying and eliminating failure modes of simpler formulations. Baseline reachability optimization ($a = \nabla \log \hat{R}$) fails under irreversible traps, high-variance stochastic transitions, and estimator corruption ($\hat{R} \neq R$). Weighting by Π suppresses irreversible collapse, variance penalization introduces risk sensitivity, and grounding via Θ reduces reliance on inconsistent estimates. In multi-agent settings, adaptive trust prevents persistent manipulation.

Across a sequence of minimal environments—featuring irreversible traps, stochastic risk, deceptive signals, and interacting agents—we show that each term removes a distinct failure mode, and that their composition yields agents that avoid irreversible collapse, reject high-risk trajectories, recover from estimator corruption, and dynamically regulate trust.

These results support a view of robust agency as the maximization of recoverable, low-uncertainty, reality-aligned future controllability, rather than unconstrained objectives.

1 Introduction

1.1 A Constrained Reachability Control Law

We introduce a control law for agents operating under irreversibility, uncertainty, and adversarial information:

$$a_t = \nabla \log (\mathbb{E}[R \cdot \Pi] - \lambda \cdot \text{Var}(R \cdot \Pi)) \cdot \Theta. \tag{1}$$

Here, R measures future controllability, Π weights recoverability of future states, λ penalizes outcome variance, and Θ modulates epistemic consistency; externally sourced estimates are incorporated via adaptive trust. We show that each term corresponds to a necessary constraint on agent behavior, and that their composition yields agents that remain robust across a range of failure modes including irreversible collapse, high-risk stochastic transitions, estimator corruption, and multi-agent manipulation.

1.2 Motivation

Agents in realistic environments must act under multiple interacting constraints. Actions may be irreversible, outcomes uncertain, and information incomplete or adversarial. Standard formulations typically address these challenges in isolation. Reinforcement learning optimizes reward but is sensitive to misspecification; empowerment-based approaches maximize influence over future states but do not distinguish recoverable from irreversible futures; probabilistic inference frameworks incorporate uncertainty but rely on prior assumptions that may not hold under adversarial conditions.

In practice, failures often arise not from insufficient optimization, but from missing constraints. Agents may enter states that irreversibly reduce future options, accept high-variance trajectories with catastrophic downside, or act on corrupted estimates of their environment. In multi-agent settings, these effects are amplified by the possibility of strategic manipulation.

1.3 Derivation Strategy

Rather than postulating a modified objective directly, we derive the proposed control law by progressively eliminating these failure modes. Starting from a baseline reachability objective,

$$a_t = \nabla \log \hat{R}(s_t), \quad (2)$$

we introduce a sequence of constraints, each motivated by a specific deficiency:

- weighting future states by reversibility to prevent irreversible collapse,
- penalizing outcome variance to suppress high-risk behavior,
- grounding estimates in observed transitions to mitigate estimator corruption, and
- adapting trust in external signals to limit sustained manipulation.

Each modification removes a distinct failure mode; their composition yields the constrained objective above.

1.4 Positioning

The proposed framework is most closely related to empowerment-based control, but differs in explicitly incorporating reversibility, uncertainty, and epistemic reliability. Unlike reward-based approaches, it does not depend on externally specified objectives. Unlike inference-based approaches, it does not rely on fixed priors, and instead emphasizes consistency between predicted and observed outcomes.

1.5 Contributions

This work makes four contributions:

- A constrained reachability control law integrating reversibility, uncertainty, and epistemic grounding;
- A constructive derivation of this law via progressive elimination of failure modes;
- Empirical validation in minimal environments demonstrating robustness across traps, stochastic risk, deception, and multi-agent interaction;

- A unified perspective on agency as the maintenance of recoverable, low-uncertainty, reality-aligned future controllability.

1.6 Scope

Results are demonstrated in minimal environments designed to isolate specific failure modes. The goal is not to establish general optimality, but to identify a compact set of constraints sufficient to eliminate common failures of reachability-based agents in these settings.

2 Preliminaries

2.1 Environment Model

We consider agents interacting with an environment modeled as a Markov decision process (MDP) defined by a state space S , action space A , and transition dynamics $P(s' | s, a)$. At each timestep t , the agent selects an action a_t based on its current state s_t and receives the next state $s_{t+1} \sim P(\cdot | s_t, a_t)$.

We assume that the agent has access to a model of the transition dynamics, either explicitly or via sampling, enabling rollout-based estimation of future trajectories.

2.2 Reachability

We define reachability $R(s)$ as a measure of the agent’s ability to influence its future state distribution starting from state s . Intuitively, $R(s)$ captures the size or diversity of the set of states that can be reached under admissible policies within a finite horizon.

Formally, we do not fix a single definition of R , but require that it satisfy the following properties:

- **Monotonicity:** states with greater controllable future variation have higher reachability,
- **Invariance:** R is invariant to irrelevant relabeling of states,
- **Computability:** R can be approximated via finite-horizon rollouts.

In practice, we approximate $R(s)$ by sampling trajectories of length T and computing a proxy for the dispersion or volume of reachable states.

2.3 Baseline Control Policy

The baseline agent selects actions by ascending the gradient of estimated reachability:

$$a_t = \nabla \log \hat{R}(s_t), \tag{3}$$

where $\hat{R}$ denotes an estimate of reachability computed from rollouts or a learned model.

This policy favors actions that increase the diversity of reachable future states.

2.4 Log-Gradient Interpretation

Using the logarithm of reachability introduces scale invariance and emphasizes avoidance of low-reachability regions. In particular, the log transformation increases sensitivity to states where $R(s)$ is small, analogous to barrier functions in control, and reduces sensitivity to uniform scaling of R .

This formulation also yields additive structure over multiplicative changes in reachability, simplifying optimization over multi-step trajectories.

2.5 Multi-Scale Reachability

To account for different time horizons, we consider a multi-scale formulation:

$$R(s) = \sum_T w_T R_T(s), \quad (4)$$

where $R_T(s)$ denotes reachability evaluated over horizon T , and w_T are non-negative weights.

Short horizons capture immediate flexibility, while longer horizons capture strategic options. Combining multiple scales allows the agent to balance short-term maneuverability with long-term controllability.

3 Failure Modes of Unconstrained Reachability

We begin by analyzing the behavior of agents that act according to the baseline reachability objective,

$$a_t = \nabla \log \hat{R}(s_t),$$

where $\hat{R}(s)$ denotes an estimate of future controllability from state s . While this objective produces coherent behavior in simple environments, it exhibits systematic failures under common conditions involving irreversible dynamics, stochastic uncertainty, and corrupted information. These failures motivate the introduction of additional constraints in subsequent sections.

3.1 Irreversible Collapse

A fundamental limitation of unconstrained reachability arises in environments containing irreversible states. Consider regions of the state space that offer high local connectivity—i.e., many immediately reachable successor states—but from which return to the broader state space is not possible. Such regions may appear favorable under a local reachability estimate, yet entering them permanently restricts future controllability.

Under the baseline policy, agents are attracted to these regions because $\hat{R}$ reflects short-term expansion without accounting for recoverability. As a result, agents systematically enter “gilded cage” states: configurations with high apparent reachability but zero long-term flexibility.

This failure highlights the distinction between *reachable* and *recoverable* futures. A control objective based solely on reachability is insufficient to avoid irreversible collapse.

3.2 High-Variance Risk

A second failure mode arises in stochastic environments where actions lead to distributions over outcomes with significantly different long-term consequences. In such settings, an action may offer high expected reachability while carrying a substantial probability of catastrophic collapse.

Under the baseline objective, agents are insensitive to the variance of outcomes and instead follow the gradient of expected reachability. This leads to risk-seeking behavior in “noisy door” scenarios, where agents repeatedly select actions with favorable expected value despite a high probability of entering low-reachability states.

This behavior demonstrates that maximizing expected reachability alone does not account for the asymmetry between recoverable gains and irreversible losses. A robust agent must distinguish between low-risk and high-risk expansions of future controllability.

3.3 Estimator Corruption

The baseline formulation assumes that the agent’s estimate $\hat{R}(s)$ accurately reflects true reachability. In practice, this estimate may be corrupted by noise, bias, or adversarial signals.

When $\hat{R}(s) \neq R(s)$, the agent follows gradients that do not correspond to actual improvements in controllability. In extreme cases, an agent may be driven toward states that appear highly favorable under $\hat{R}$ but in fact lead to a collapse of true reachability. This corresponds to a form of wireheading, in which the agent optimizes its internal estimate rather than the underlying objective.

Importantly, this failure is not due to suboptimal optimization, but to misalignment between estimation and reality. A control law that depends directly on $\hat{R}$ without accounting for estimator reliability is therefore vulnerable to systematic error.

3.4 Social Manipulation

In multi-agent environments, agents may exchange information about the state of the environment or the consequences of actions. While such communication can improve estimation, it also introduces the possibility of strategic manipulation.

An agent that relies on externally provided estimates of reachability can be influenced by other agents that provide misleading signals. Under the baseline formulation, externally sourced estimates are treated equivalently to internally generated ones, allowing adversarial agents to steer behavior by distorting perceived gradients.

This leads to a characteristic pattern: agents may initially follow deceptive signals into low-reachability regions before correcting their behavior based on subsequent observations. While recovery may be possible, the initial vulnerability can result in irreversible loss.

This failure mode illustrates that robustness in multi-agent settings requires not only accurate estimation, but also mechanisms for evaluating the reliability of information sources.

3.5 Summary

These failure modes share a common structure: each arises from an implicit assumption in the baseline reachability objective.

- Irreversible collapse arises from ignoring recoverability,
- High-variance risk arises from ignoring outcome variability,
- Estimator corruption arises from assuming estimator accuracy,
- Social manipulation arises from treating all information sources as equally reliable.

In each case, the failure is not due to the reachability objective itself, but to the absence of constraints that account for these factors. In the next section, we introduce a sequence of modifications that address each failure mode in turn, yielding a constrained reachability formulation that remains robust under these conditions.

4 Derivation of Constrained Reachability

The failure modes identified in the previous section arise from implicit assumptions in the baseline reachability objective. We now introduce a sequence of constraints that remove these assumptions. Each modification addresses a specific failure mode, and their composition yields the control law presented in Section 1.

4.1 Reversibility Constraint

The failure of unconstrained reachability in the presence of irreversible states suggests that not all reachable futures should be treated equally. In particular, states that cannot be exited should be downweighted relative to those that preserve access to the broader state space.

We introduce a reversibility factor $\Pi(s \rightarrow s') \in [0, 1]$, which measures the degree to which future states remain recoverable. Intuitively, Π is high for states from which return to the originating region is likely, and low for states that lead to irreversible collapse. In practice, Π can be approximated via return probability under a learned or sampled policy, or via shortest-path distance in the state transition graph.

This yields a modified reachability measure:

$$R^*(s) = \mathbb{E}_{s' \sim P(\cdot|s)} [R(s') \cdot \Pi(s \rightarrow s')] . \quad (5)$$

By weighting future states according to their recoverability, this formulation suppresses trajectories that lead into irreversible traps while preserving those that maintain long-term controllability.

4.2 Uncertainty Constraint

While reversibility addresses irreversible collapse, it does not account for stochastic variability in outcomes. As shown in Section 3, actions with high expected reachability may still be undesirable if they carry a significant probability of catastrophic outcomes.

To address this, we introduce a variance penalty over the reversibility-weighted reachability:

$$R^{**}(s) = \mathbb{E}[R(s') \cdot \Pi(s \rightarrow s')] - \lambda \cdot \text{Var}(R(s') \cdot \Pi(s \rightarrow s')), \quad (6)$$

where $\lambda \geq 0$ controls sensitivity to risk.

This modification discourages trajectories with high outcome variance, thereby reducing the tendency to pursue high-risk, high-reward strategies that may lead to irreversible loss. The resulting objective distinguishes between stable expansions of controllability and those that are dominated by rare but severe failures.

4.3 Epistemic Grounding

The preceding formulation assumes that the agent’s estimate of reachability is accurate. However, as discussed in Section 3, estimator corruption can lead to systematic divergence between perceived and actual controllability.

To mitigate this, we introduce an epistemic grounding factor $\Theta \in [0, 1]$ that modulates the influence of the agent’s internal estimates based on their consistency with observed outcomes. Let ϵ_t denote a measure of prediction error over recent timesteps. We define:

$$\Theta = f(\epsilon_{t-k:t}), \quad (7)$$

where f is a decreasing function of prediction error (e.g., an exponential decay).

This factor reduces the impact of internally generated estimates when they are inconsistent with observed transitions, effectively downweighting unreliable models. The resulting control signal becomes:

$$\hat{R}_{\text{grounded}}^{**} = \Theta \cdot \hat{R}^{**}. \quad (8)$$

Grounding ensures that the agent’s behavior remains anchored to empirical feedback, preventing persistent divergence due to estimator bias or corruption.

4.4 Adaptive Trust

In multi-agent settings, agents may incorporate information from external sources. While such signals can improve estimation, they also introduce vulnerability to manipulation.

We model external inputs as signals $\sigma_{j \rightarrow i}$ provided by agent j to agent i , and introduce trust weights $w_{ji} \in [0, 1]$ that determine their influence. Trust is updated based on the consistency between received signals and observed outcomes. For example, a simple update rule takes the form:

$$w_{ji}(t+1) = w_{ji}(t) \cdot \exp(-\gamma \cdot \epsilon_{ji}(t)), \quad (9)$$

where $\epsilon_{ji}(t)$ measures the discrepancy between predicted and realized outcomes, and γ controls the rate of decay.

External estimates are then incorporated via a weighted aggregation:

$$\hat{R}^{\text{social}} = \sum_j w_{ji} \cdot \sigma_{j \rightarrow i}. \quad (10)$$

This mechanism allows agents to benefit from reliable information sources while reducing the influence of inconsistent or adversarial signals. We emphasize that trust dynamics are adaptive and may not converge in all environments; however, they enable recovery from persistent manipulation in the settings considered here.

4.5 Unified Objective

Combining the above components yields a grounded, risk-aware, and reversibility-constrained estimate of reachability. The resulting control law can be expressed as:

$$a_t = \nabla \log (\mathbb{E}[R \cdot \Pi] - \lambda \cdot \text{Var}(R \cdot \Pi)) \cdot \Theta, \quad (11)$$

with externally sourced estimates incorporated via trust-weighted aggregation.

This formulation integrates all constraints introduced above. Each term plays a distinct role: Π suppresses irreversible collapse, the variance term penalizes high-risk trajectories, Θ enforces consistency with observed outcomes, and trust weighting regulates the influence of external information.

We do not claim that this formulation is unique; rather, we show that it constitutes a minimal composition sufficient to eliminate the failure modes identified in Section 3 within the environments considered.

5 Experimental Setup

We evaluate the constrained reachability framework using a sequence of minimal environments designed to isolate the failure modes identified in Section 3. Each environment introduces a specific structural challenge—irreversibility, stochastic risk, estimator corruption, or multi-agent interaction—allowing direct comparison between baseline and constrained agents.

5.1 Environment Classes

We consider the following classes of environments:

Irreversible Trap Environments. These environments contain regions with high local connectivity but no return path to the broader state space. Such regions act as absorbing sets, capturing agents that enter them. These are used to evaluate sensitivity to irreversible collapse.

Stochastic Transition Environments. Actions in these environments lead to probabilistic outcomes with significantly different long-term consequences. In particular, certain transitions offer high expected reachability but include a substantial probability of entering low-reachability states. These environments test sensitivity to outcome variance.

Estimator Corruption Environments. Agents receive perturbed or adversarially modified estimates of reachability. This is implemented by introducing bias or noise into the agent’s internal model or by injecting misleading signals into its observations. These environments test robustness to estimator-reality mismatch.

Multi-Agent Environments. Multiple agents share a common state space and may exchange information about the environment. Agents can influence one another through signals that affect perceived reachability. These environments test susceptibility to manipulation and the effectiveness of adaptive trust.

All environments are implemented either as discrete state-transition graphs or low-dimensional continuous spaces with locally defined dynamics. Transitions are either deterministic or stochastic, depending on the setting.

5.2 Agent Variants

We evaluate a sequence of agents corresponding to progressively constrained formulations:

- **Baseline (R):** agents act according to $a = \nabla \log \hat{R}$,
- **+ Reversibility (R*):** reachability weighted by Π ,
- **+ Uncertainty (R**):** variance-penalized objective,
- **+ Grounding:** modulation via Θ ,
- **+ Trust:** incorporation of external signals via adaptive weights w_{ji} .

Each variant differs only in the objective used to evaluate candidate actions; all other aspects of the agent (state representation, rollout procedure, and update rules) are held constant.

5.3 Reachability Estimation

Reachability is estimated via finite-horizon rollouts. For each candidate action, the agent simulates trajectories up to a fixed horizon T and computes a proxy for controllability based on the dispersion or volume of reachable states.

In discrete environments, this corresponds to the number of distinct states reachable within the horizon. In continuous environments, it is approximated via the spread of sampled trajectories.

Reversibility Π is estimated using either:

- empirical return probability to the originating region under sampled trajectories, or
- shortest-path distance in the state-transition graph, mapped to $[0, 1]$ via an exponential decay.

Variance is computed over rollout outcomes, and grounding Θ is derived from recent prediction error between expected and observed transitions.

5.4 Metrics

We evaluate agent behavior using the following metrics:

- **Trap Entry Rate:** fraction of runs in which the agent enters an irreversible region,
- **Survival Time:** number of steps before entering low-reachability states,
- **Estimator Error:** deviation between estimated and realized reachability,
- **Recovery Time:** time required to return to high-reachability regions after perturbation,
- **Trust Dynamics:** evolution of weights w_{ji} in multi-agent settings.

These metrics are chosen to directly reflect the failure modes identified in Section 3.

5.5 Evaluation Protocol

For each environment and agent variant, we run multiple independent trials with identical initial conditions and randomized stochastic transitions where applicable. Performance is reported as averages over runs.

Comparisons are made incrementally, evaluating the effect of adding each constraint relative to the preceding formulation. This allows isolation of the contribution of each component in addressing its corresponding failure mode.

No external reward signals are used; all behavior arises from the reachability-based objectives defined in Section 4.

6 Results

We evaluate the constrained reachability framework across the environments described in Section 5, comparing baseline and progressively constrained agents. Results are organized according to the failure modes identified in Section 3. In each case, we isolate the effect of introducing the corresponding constraint.

6.1 Elimination of Irreversible Collapse

In irreversible trap environments, baseline agents consistently enter high-connectivity regions that lack return paths, resulting in permanent loss of access to the broader state space. This behavior reflects the inability of unconstrained reachability to distinguish between reachable and recoverable futures.

Introducing the reversibility constraint (II) eliminates this failure mode. Agents using R^* avoid entering trap regions, instead selecting trajectories that preserve long-term access to alternative states. Across runs, trap entry rates decrease substantially relative to baseline, and survival time increases correspondingly.

These results demonstrate that weighting future states by recoverability is sufficient to prevent irreversible collapse in the environments considered.

6.2 Suppression of High-Risk Behavior

In stochastic transition environments, baseline and reversibility-constrained agents (R and R^*) both exhibit risk-seeking behavior, repeatedly selecting actions with high expected reachability despite significant probability of collapse.

Adding the uncertainty constraint (R^{**}) alters this behavior. Agents become sensitive to outcome variance and preferentially select lower-risk trajectories, even when these have lower expected reachability. This leads to a marked reduction in catastrophic outcomes and increased stability across runs.

These results indicate that reversibility alone is insufficient under stochastic dynamics, and that variance penalization is necessary to suppress high-risk strategies.

6.3 Recovery from Estimator Corruption

In environments with corrupted or adversarially perturbed reachability estimates, baseline agents follow misleading gradients, often entering low-reachability states that appear favorable under $\hat{R}$. This produces a divergence between estimated and realized controllability.

Incorporating epistemic grounding (Θ) mitigates this effect. When prediction error increases, the influence of inconsistent estimates is reduced, allowing the agent to re-anchor its behavior to observed transitions. Empirically, agents exhibit transient degradation followed by recovery, returning to high-reachability regions after periods of estimator mismatch.

This behavior contrasts with baseline agents, which remain trapped under persistent estimator corruption. The results show that grounding enables recovery from, though not complete prevention of, deception.

6.4 Robustness under Social Interaction

In multi-agent environments, agents that incorporate external signals without filtering are susceptible to manipulation. Baseline agents follow misleading inputs from other agents, leading to suboptimal or irreversible outcomes.

Introducing adaptive trust weighting reduces this vulnerability. Trust in external signals decreases when predictions derived from those signals fail to match observed outcomes. As a result, agents initially influenced by deceptive signals subsequently downweight unreliable sources and revert to behavior guided by grounded internal estimates.

Across runs, this produces a characteristic pattern: initial susceptibility followed by rapid recovery. While agents may be temporarily misled, persistent control by adversarial signals is not observed in the evaluated settings.

6.5 Summary

Across all environments, each constraint addresses a distinct failure mode:

- Reversibility (Π) prevents irreversible collapse,
- Variance penalization (λVar) suppresses high-risk behavior,
- Grounding (Θ) enables recovery from estimator corruption,
- Adaptive trust limits sustained social manipulation.

The full formulation, combining all components, exhibits robust behavior across all tested conditions. We emphasize that robustness emerges from the interaction of these constraints; no single component is sufficient to address all failure modes.

7 Discussion

7.1 Minimal Constraints for Robust Agency

The results suggest that robust behavior does not arise from unconstrained optimization of reachability, but from the integration of a small set of additional constraints. Each component introduced in Section 4 addresses a distinct failure mode identified in Section 3, and their combination yields agents that remain stable across all tested environments.

Importantly, these components are not arbitrary augmentations. Each is introduced as a necessary modification to eliminate a specific class of failure. Reversibility distinguishes recoverable from irreversible futures, variance penalization distinguishes stable from high-risk trajectories, grounding distinguishes consistent from unreliable estimates, and adaptive trust distinguishes reliable from adversarial information sources.

We do not claim that the resulting formulation is unique. Rather, the empirical results support a minimality claim: the combination of these constraints is sufficient to eliminate the observed failure modes in the environments considered, and removing any component reintroduces a corresponding failure.

7.2 Relation to Existing Work

The proposed framework is most closely related to empowerment-based approaches, which also emphasize control over future states. However, standard empowerment formulations do not explicitly account for irreversibility or outcome variance, and therefore exhibit the failure modes described in Section 3.

Reward-based reinforcement learning provides a general framework for optimization, but relies on externally specified objectives and is sensitive to misspecification. In contrast, the reachability-based formulation defines an intrinsic objective grounded in controllability, while the introduced constraints address structural limitations rather than reward design.

Active inference and related probabilistic frameworks emphasize inference over latent states and policies, typically under fixed prior assumptions. While these approaches incorporate uncertainty, they do not explicitly distinguish between reversible and irreversible dynamics or between reliable and adversarial information sources.

Table 1 summarizes these distinctions.

Framework	Limitation Addressed Here
Reinforcement Learning	Reward misspecification; lacks intrinsic controllability objective
Empowerment	Does not account for irreversibility or risk
Active Inference	Depends on priors; lacks explicit controllability objective

Table 1: Comparison with related frameworks.

7.3 Implications for Alignment

The results have implications for the design of robust agents in safety-critical settings. In particular, they suggest that failure modes commonly associated with misalignment—such as irreversible side effects, risk-seeking behavior, and susceptibility to deceptive inputs—can arise from structural deficiencies in the control objective rather than from specific reward functions.

From this perspective, alignment is not solely a matter of specifying correct objectives, but of ensuring that agents operate under constraints that preserve recoverability, limit exposure to catastrophic uncertainty, and maintain consistency between internal estimates and external reality. The introduction of adaptive trust further highlights the importance of epistemic robustness in multi-agent settings.

These observations support a layered view of alignment in which robustness emerges from the interaction of multiple constraints, rather than from a single optimization criterion.

7.4 Stability of Grounding and Trust

The grounding and trust mechanisms introduced in Section 4 are adaptive and depend on observed prediction error. While empirical results show that these mechanisms enable recovery from estimator corruption and adversarial signals, they do not guarantee convergence in all environments.

In particular, trust weights may oscillate in settings with rapidly changing or strategically adaptive agents, and grounding may suppress useful exploratory behavior if prediction error is persistently high. These dynamics suggest a tradeoff between responsiveness and stability that is not fully characterized in the present work.

Further analysis is required to establish conditions under which trust and grounding converge, as well as to understand their behavior in more complex multi-agent systems.

7.5 Limitations

This work is subject to several limitations.

First, all results are obtained in minimal environments designed to isolate specific failure modes. While this allows clear causal attribution, it does not establish scalability to high-dimensional or real-world domains.

Second, reachability and reversibility are approximated via finite-horizon rollouts and heuristic measures. More precise or efficient estimation methods may be required in complex environments.

Third, trust is initialized and updated heuristically, and the problem of trust bootstrapping—how agents form reliable initial estimates of other agents’ credibility—is not addressed.

Finally, while the proposed formulation eliminates the failure modes considered here, it may not account for other classes of failure that arise in more complex settings.

7.6 Summary

Taken together, these results support a view of agency as the maintenance of controllable, recoverable futures under uncertainty and imperfect information. The constrained reachability formulation provides a compact expression of this principle, while the staged derivation clarifies the role of each component in ensuring robustness.

8 Conclusion

We introduced a constrained reachability control law for agents operating under irreversibility, uncertainty, and adversarial information, and derived it by progressively eliminating failure modes of unconstrained reachability. The resulting formulation integrates reversibility weighting, variance penalization, epistemic grounding, and adaptive trust into a single objective governing action selection.

Across a sequence of minimal environments, we showed that each component addresses a distinct and necessary constraint on agent behavior. Reversibility prevents irreversible collapse, variance penalization suppresses high-risk trajectories, grounding enables recovery from estimator corruption, and adaptive trust limits sustained manipulation in multi-agent settings. Their composition yields agents that remain robust across all tested conditions.

These results suggest that robust agency is not captured by unconstrained optimization of future controllability, but requires a structured integration of constraints that account for temporal structure, uncertainty, and epistemic reliability. Rather than proposing a universal solution, this work identifies a compact set of mechanisms that together eliminate a range of common failure modes.

Future work includes scaling the framework to higher-dimensional environments, improving estimation of reachability and reversibility, and analyzing the dynamics of trust and grounding in larger multi-agent systems. More broadly, the approach suggests a pathway toward designing agents whose behavior remains stable under conditions that routinely induce failure in simpler formulations.

A Formal Definitions

B Additional Experiments

C Implementation Details